

Rohit Annasaheb Ragde

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PROFESSIONAL SUMMARY

Finance-domain Data Engineer building AML transaction monitoring pipelines, regulatory reporting systems, and cloud data infrastructure at TD Bank — with 7 years of experience architecting enterprise-scale ETL/ELT pipelines, lakehouse architectures, and analytics platforms across financial services, utilities, and technology. Deep engineering expertise in PySpark, dbt, Snowflake, Airflow, Azure Databricks, and AWS — with a strong track record automating high-volume workflows, tuning pipeline performance at scale, and migrating data infrastructure to cloud. Analytics capabilities in Power BI and Tableau complement core engineering strengths — translating complex financial datasets into compliance and executive dashboards where needed.

TECHNICAL SKILLS

Languages & Scripting: Python, SQL, T-SQL, PL/SQL, Bash, Unix Shell Scripting (advanced — pipeline automation, batch job scheduling, failure handling, cron)

Data Engineering: ETL/ELT Pipelines, Data Modeling, CDC, SCD, Data Quality & Validation, Schema Design, Dimensional Modeling

Analytics & BI: Power BI, Tableau, KPI Dashboards, Financial Reporting, AML Analytics, Fraud Detection, Variance Analysis, Data Storytelling, Self-Serve Analytics, Cross-functional Stakeholder Management

Big Data & Streaming: PySpark, Apache Spark, Apache Kafka (familiar), Spark Streaming, Delta Lake

Cloud — AWS: S3, RDS, Glue, EMR, Lambda, MWAA (Managed Airflow), Redshift (familiar), IAM, CloudWatch

Cloud — Azure: Azure Databricks, Azure Data Lake Gen2, Azure Data Factory, Azure Synapse Analytics, Azure SQL Database

Cloud — GCP: Google Cloud Platform, Cloud Storage, BigQuery (familiar), Cloud Composer

Cloud — Snowflake & Other: Snowflake (Warehouses, Stages, Streams, Tasks, dbt integration), Teradata Vantage Cloud, Oracle Cloud Infrastructure (OCI)

Orchestration & Transform: Apache Airflow, dbt (models, tests, snapshots, macros), Azure Data Factory

Databases: SQL Server, PostgreSQL, Oracle, MongoDB, Teradata

DevOps & CI/CD: Git, GitHub, GitHub Actions (workflow automation, CI/CD pipelines, scheduled jobs), Docker, Jenkins, CI/CD

GenAI & AI Engineering: Claude API, Prompt Engineering, Multi-turn Conversational Workflows, API-driven AI Agent Orchestration

Security & Compliance: TDE, RBAC, Auditing, Data Masking, GDPR, HIPAA, SOX, AML/SAR/CTR Regulatory Frameworks

PROFESSIONAL EXPERIENCE

Data Engineer

Oct 2025 – Present

TD Bank — Financial Services (via Protiviti)

Philadelphia, PA

- Architected and maintained scalable SQL-based data pipelines integrating 15+ source systems powering AML transaction monitoring, regulatory reporting, and financial crime analytics for one of Canada's largest banks
- Leading migration of analytics infrastructure to AWS — designing S3 data lake architecture, RDS as relational database layer, PySpark for large-scale transformations, and Airflow for pipeline orchestration
- Engineered 30+ automated data quality validation frameworks using Python and SQL — null checks, reconciliation logic, and anomaly detection — reducing data incidents by 40% and ensuring SOX compliance
- Automated manual reconciliation and reporting workflows saving 40+ hours/month — leveraging Python, Bash shell scripting, and GitHub Actions CI/CD pipelines to eliminate manual intervention across financial data processes
- Performed large-scale variance analysis and data reconciliation across financial datasets, identifying systemic discrepancies and improving downstream regulatory reporting reliability
- Engaged with finance leadership and compliance stakeholders to translate regulatory and analytical requirements into scalable data engineering solutions, serving as primary data liaison for AML reporting
- Designed Power BI and Tableau dashboards tracking 20+ financial KPIs and AML metrics — consumed daily by compliance leadership and finance stakeholders to support data-driven decisions

Data Engineer

Sep 2024 – Sep 2025

Cantonica (StableCoupons Inc) — Technology Startup

Philadelphia, PA

- Owned end-to-end data engineering and analytics stack — designing, building, and maintaining the full data pipeline infrastructure from ingestion through transformation to BI reporting, functioning as the primary data engineer and analyst for the product
- Designed and built end-to-end ELT data pipelines using Snowflake and dbt on GCP infrastructure, implementing structured dimensional data models and schema design that reduced downstream data preparation time by 30%
- Optimized complex SQL queries and Snowflake schema architecture improving data processing performance by 25% — profiling execution plans, restructuring joins, and applying clustering keys on high-velocity datasets

- Leveraged Google Cloud Platform (GCP) services for scalable data storage, compute, and pipeline execution — integrating cloud infrastructure with Snowflake and dbt transformation workflows
- Implemented robust automated data quality and validation frameworks (null checks, reconciliation logic, monitoring alerts) ensuring pipeline reliability, data accuracy, and completeness across production workloads
- Integrated data pipelines with Django application backend using MongoDB and REST APIs, enabling real-time data flow between engineering systems and product-facing analytics features
- Delivered Power BI and Tableau dashboards enabling real-time product usage analytics — partnering with product and operations stakeholders to translate business requirements into reporting solutions
- Documented data pipeline architecture, quality standards, and analytical best practices, improving team knowledge and reducing onboarding time for engineering workflows

Data Engineer

Jun 2023 – Dec 2023

Exelon Inc — Utility Services

Wilmington, DE

- Built and optimized high-volume PySpark pipelines on Azure Databricks processing 15+ TB of daily utility operations data — applying Delta Lake for ACID-compliant storage, partitioning strategies, broadcast joins, and caching optimizations to achieve 95% SLA reliability
- Engineered Azure Data Factory workflows and Python-based ETL transformations migrating enterprise data from on-premises SQL Server to Azure Synapse Analytics, reducing data latency by 45%
- Reduced critical pipeline execution time by 30% through systematic performance tuning — profiling bottlenecks, restructuring transformation logic, and implementing indexing strategies across 20+ enterprise tables
- Developed scalable PySpark transformation pipelines for 10+ datasets applying business rules, schema normalization, and data standardization across hybrid on-premises and cloud environments
- Built comprehensive automated data quality validation frameworks using Python, SQL, and Bash shell scripts — reducing production data anomalies by 40% and establishing real-time alerting systems for pipeline failures

Data Engineer

Sep 2019 – Aug 2022

Teradata Inc — Enterprise Data Consulting (Clients: Audi, Bank of Ireland, Volkswagen)

Mumbai, India

- Optimized 100+ complex SQL queries across enterprise financial and automotive clients on Teradata Vantage Cloud — using execution plan analysis, indexing strategies, partitioning, and stored procedure refactoring to transform query response times from minutes to seconds
- Architected and maintained 10+ enterprise ETL/ELT pipelines using Teradata Vantage, Talend, and SQL Server across retail, manufacturing, and telecom clients — processing 2.5TB daily with 95% pipeline reliability
- Built 40+ Python and Bash shell scripts for pipeline execution monitoring, failure detection, cron-based scheduling, and automated recovery — reducing mean time to resolution from hours to minutes across production workloads
- Led development of SLIPSTREAM — an automated ELT pipeline on Teradata Vantage Cloud processing 10M+ telecom subscriber records daily with cron-based scheduling, improving revenue forecasting accuracy by 25%
- Designed dimensional data models and implemented workload management strategies improving system throughput and query scheduling stability across multi-client Teradata Vantage environments
- Developed inventory optimization and customer recommendation systems processing 20M+ retail transactions using Teradata and Python, delivering dynamic pricing solutions that increased client revenue by 15%

PROJECTS

Real-Time Financial Data Streaming Pipeline

Jun 2026 – Present

Apache Kafka · AWS · Snowpipe · Snowflake Streams & Tasks · Python

- Building a real-time financial transaction streaming pipeline using Apache Kafka as the core event streaming platform — ingesting and processing 1M+ daily financial transactions with low-latency, fault-tolerant message delivery
- Architecting end-to-end streaming architecture: Kafka producers publishing financial events → AWS-hosted Kafka brokers → Snowpipe for continuous auto-ingestion into Snowflake → Streams and Tasks for incremental transformations and downstream analytics
- Implementing exactly-once semantics, consumer group management, and topic partitioning strategies to ensure reliable, ordered processing of high-volume financial transaction data
- Engineered a multi-agent AI system using Claude API featuring 6 specialized agents with multi-turn conversational workflows, real-time data retrieval, and automated pipeline orchestration
- Implemented prompt engineering patterns, JSON/API response transformation, and metadata-driven architecture integrating 4 external APIs (Anthropic, Happenstance, Gmail, Greenhouse) with automated fallback logic
- Built API-driven AI workflows with structured error handling, SQLite persistence layer, and GitHub Actions automation — demonstrating production-grade GenAI system design

Scalable Financial Transactions Data Pipeline

Mar 2026 – Apr 2026

PySpark · dbt · Airflow · SQL · Delta Lake

- Designed end-to-end pipeline ingesting and processing financial transaction data simulating real-world banking analytics — fact/dimension modeling (transactions, users, merchants) with PySpark transformations and dbt
- Orchestrated pipeline execution using Airflow with automated validation frameworks (null handling, deduplication, anomaly detection) ensuring reliable financial reporting
- Implemented Delta Lake for ACID-compliant storage and incremental data processing, enabling reliable financial data versioning and audit trail support

NYC Taxi Analytics Pipeline — Medallion Architecture

Feb 2026 – Mar 2026

Apache Spark · Delta Lake · Python · Azure Databricks

- Built Bronze/Silver/Gold medallion lakehouse architecture processing high-volume NYC taxi datasets with Spark-based transformations, partitioning strategies, and incremental data processing on Azure Databricks
- Implemented schema enforcement, data quality monitoring, and anomaly detection enabling business insights into trip patterns, peak demand, and revenue drivers
- Leveraged Delta Lake for reliable, versioned data storage across medallion layers — demonstrating production-grade lakehouse design patterns applicable to enterprise financial data platforms

EDUCATION

Master of Science, Computer Science

Aug 2022 – Jun 2024

Drexel University

Philadelphia, PA

Coursework in distributed systems, big data, machine learning, and cloud computing

Bachelor of Engineering, Computer Engineering

Jul 2016 – May 2019

University of Mumbai

Mumbai, India

CERTIFICATIONS

- Teradata Database Vantage Certified Administrator — Teradata (Jun 2021)
- Teradata Database Vantage Certified SQL Associate — Teradata (Mar 2021)
- Oracle Database SQL 1Z0-071 Certified SQL Associate — Oracle (Aug 2018)